

Attention to Privacy Sensitive Information in Input Text

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<https://github.com/pals-ucb/privacy-ner-att>

Abstract

In this work, we present a novel approach for detecting privacy-sensitive information in text by extending a transformer-based encoder foundational model (Modern BERT) with a specialized cross attention mechanism that attends personal names to PII identifiers detected through named entity recognition (NER) components to classify the input text. The problem of privacy detection in text involves identifying instances where personal identifiers (e.g., names) are linked with sensitive attributes (e.g., social security numbers or medical conditions). Our approach uses a multi-layered architecture combining NER and attention mechanisms to correlate such identifiers with potential private information. We hypothesize that an NER-driven attention strategy can effectively highlight these correlations, improving the accuracy of privacy disclosure detection. We design a foundational model with two NER tagging heads (PERSON and PII tags) and a cross-attention layer that feeds into a final privacy classifier. The model is trained on a synthetic dataset with token-level ONTO Notes BIO annotations and a privacy classification label. Experimental results show that this multi-task architecture achieves high accuracy in classifying sensitive texts, while an ablation study confirms the benefit of the dual NER heads and cross-attention. We also survey related works – including traditional rule-based systems, NER-augmented transformers, and layered models for PII detection – to position our contributions. The proposed model lays the groundwork for more context-aware privacy detection in textual data.

1.0 Introduction

1.1 Problem Statement

Privacy-sensitive information in text refers to content that can reveal personal details about an individual, such as identity or private attributes. Automatically detecting such information is crucial for data privacy, compliance (e.g., GDPR), and preventing inadvertent disclosures. The task is challenging because sensitive disclosures often arise from specific combinations of tokens – for example, a person’s name appearing alongside a Social Security Number (SSN) or a health condition in the same context. Simple keyword spotting is insufficient; effective detection requires understanding context and relationships between entities.

1.2 Named Entity Recognition for Privacy

A promising approach is to leverage Named Entity Recognition (NER) to identify key tokens (e.g., PERSON names, dates, IDs) and then determine if their context implies a privacy risk. We explore an NER-based method with attention mechanisms that link personal identifiers to potentially sensitive information. The central hypothesis is that NER-based attention can effectively correlate personal identifiers (like a PERSON name) with privacy-sensitive tokens (like an SSN or medical CONDITION), thereby flagging text that contains personal disclosures. By focusing on entity interactions, the model can

learn patterns such as “if a Person’s name is associated through some relation to PII sensitive tokens like SSN, CC, Health condition, etc., then this is likely sensitive.” This approach differs from classic classification in that it explicitly uses token-level entity information as an intermediate step to inform the final decision.

1.3 Related Work

Early methods for detecting personal or sensitive information relied heavily on handcrafted rules and pattern matching. For example, de-identification systems in clinical text (e.g., in the MIMIC medical database) used extensive dictionaries and regular expressions to mask Protected Health Information (PHI). Such rule-based systems (e.g., Philter-UCSF) can achieve high precision but require significant manual effort and may not generalize well. Machine learning techniques introduced statistical modeling for PII detection: classifiers like Support Vector Machines and Conditional Random Fields (CRFs) were used to identify entity spans corresponding to PII categories. With the rise of deep learning, researchers began using neural networks for this task. In recent years, transformer-based models have become prevalent for sensitive information detection. BERT and its variants have been fine-tuned for de-identification tasks in domains like healthcare, demonstrating state-of-the-art performance in identifying names, dates, and other PHI in text. Johnson et al. (2020) developed one of the early transformer models for clinical note de-identification, using a bidirectional transformer to identify PHI while addressing data scarcity through pretraining. Similarly, Wang et al. (2022) combined a BERT-based encoder with a CRF decoder to improve PHI. These works show that transformer encoders, often augmented with sequence labeling layers like CRFs, provide strong baselines for PII/PHI identification. Another relevant line of research is layered and multi-task architectures for PII detection. Jian et al. (2017) proposed a cascaded system that first applied rule-based filters then machine learning for PHI, effectively layering simple and complex detectors. More recently, researchers have explored NER-augmented language models and multi-task transformers. For example, Mo et al. (2023) introduced a multi-task Transformer that incorporates an auxiliary entity-boundary detection task into the NER process, using cross-attention to integrate external knowledge about entity types. This indicates the promise of cross-attention mechanisms in fusing information from multiple sources or tasks to improve entity-related predictions. Our approach draws inspiration from these multi-task models: we extend a transformer with dual NER heads and a cross-attention layer, hypothesizing that this layered design is privacy-aware – i.e., it is explicitly tuned to catch interactions between different entity types that signal a privacy disclosure.

In summary, prior work demonstrates that: (a) leveraging linguistic entities is beneficial for detecting private information; (b) transformer-based models with sequence tagging are effective for PII/PHI recognition; and (c) multi-task or layered

approaches can enhance detection by combining signals (rules + ML, or multiple NER tasks). Building on these insights, we propose an extended Modern BERT model that uses two NER tagging layers and cross-attention to classify text for privacy-sensitive content. We next describe our approach, including the model architecture and the data strategy, followed by experimental validation.

2.0 Approach

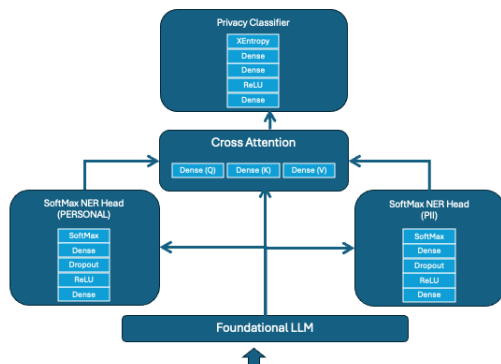
2.1 Objective

The primary objective of this research is to develop and evaluate a specialized transformer-based model capable of accurately identifying and classifying personally identifiable information (PII) and sensitive personal entities within unstructured textual data. The project aims to address critical gaps in existing privacy-preserving approaches by incorporating advanced Named Entity Recognition (NER) and cross-attention mechanisms, thereby improving detection performance and interpretability.

2.2 Model Architecture

Our architecture is a multi-layered extension of a transformer encoder-based Foundational Model that integrates token-level entity recognition with document-level classification. Figure 1 illustrates the overall design

Figure 1: Privacy NER Attention to PII Model



In Figure 1, the base transformer encoder Foundational Model produces contextualized embeddings for each token in the input text. On top of this encoder, we attach two parallel NER heads: one dedicated to recognizing PERSON entities (names of people) and another dedicated to identifying PII entities (other sensitive tokens such as IDs, account numbers, conditions, etc.). Each NER head can be implemented as a feed-forward tag predictor or enhanced with a CRF layer for structured prediction. The outputs of these two NER streams are then fed into a cross-attention layer that allows the model to attend to interactions between person names and other sensitive tokens. The cross-attention produces a fused representation, which is finally passed to a binary classification head that predicts whether the entire input text contains a privacy-sensitive disclosure. By splitting the entity recognition into two specialized heads, we encourage the model to learn a rich

representation of each entity type, and by using cross-attention, we enable it to learn relationships between the two types of entities of co-occurrence (e.g., how likely a PERSON tag associated through a relation to a nearby PII tag constitutes a privacy violation).

2.3 Modeling Decisions

A key design decision was to use two distinct NER heads rather than a single combined tagger. This allows the model to treat personal names and other private information differently. Names are often common nouns that rely on context to identify, whereas sensitive tokens like SSNs or medical terms have distinct formats or vocabularies. By training separate taggers, the model can optimize for each sub-problem (e.g., better recall on names, and precise patterns for IDs) without interference. Another decision was to introduce a cross-attention mechanism between these heads. We considered simpler concatenation or feature fusion, but cross-attention offers a learnable way to highlight relevant connections: the PERSON head's output can attend to tokens flagged by the PII head and vice versa, effectively learning which person-token and sensitive-token pairs are important. This is akin to an interactive layer that simulates how a human reader might connect "John Doe" with "123-45-6789" in the sentence "John Doe's SSN is 123-45-6789." The cross-attention layer, implemented as multi-head attention blocks taking queries from one head and keys/values from the other, yields context-enriched representations that inform the final classifier.

2.4 Datasets Evaluated

Finding the right dataset for this task was challenging, as we needed text annotated for both named entities and overall privacy content. We initially explored existing NER datasets and general-purpose corpora:

- CoNLL-2003: A widely used NER dataset (news domain) with person, organization, location, and miscellaneous entity tags. We considered leveraging it for person name tagging. However, CoNLL-2003 lacks annotations for sensitive personal data (e.g., no SSNs or medical conditions) and does not directly indicate which sentences are privacy-sensitive. It is domain-mismatched for our purposes, so we ruled it out.
- ONTO Notes 5.0: A rich corpus with multiple genres and a variety of entity types (including PERSON, ORG, GPE, etc.). ONTO Notes uses a BIO tagging format, which aligns with our needs. Yet, it still does not label things like SSNs or medical terms as special entities (since those are not standard named entities in ONTO Notes). We took inspiration from its tagging format but not its content.

Given the lack of an existing dataset that directly matches our problem (paired person-PII annotations and privacy labels), we opted to create a synthetic dataset. This dataset generation was a major component of our approach, designed to provide both token-level and sequence-level supervision. The dataset creation process is described later in this paper.

2.5 The Attention Mechanism

Consider the following sensitive sentence as an example:

“John Adams was admitted to Saint Mary’s Hospital for drug addiction treatment. During admission, John’s personal information, including his social security number (SSN: 123-45-6789) and credit card number (CC: 4111-1111-1111), was recorded and inadvertently made publicly available.”

Using ONTO Notes BIO tagging, the sentence is annotated as follows:

[(‘John’, ‘B-PERSON’), (‘Adams’, ‘I-PERSON’), (‘was’, ‘O’), (‘admitted’, ‘O’), (‘to’, ‘O’), (‘Saint’, ‘B-ORG’), (‘Mary’, ‘I-ORG’), (‘’s’, ‘I-ORG’), (‘Hospital’, ‘I-ORG’), (‘for’, ‘O’), (‘drug’, ‘B-CONDITION’), (‘addiction’, ‘I-CONDITION’), (‘treatment’, ‘O’), (‘.’, ‘O’), (‘During’, ‘O’), (‘admission’, ‘O’), (‘,’, ‘O’), (‘John’, ‘B-PERSON’), (‘’s’, ‘O’), (‘personal’, ‘O’), (‘information’, ‘O’), (‘,’, ‘O’), (‘including’, ‘O’), (‘his’, ‘O’), (‘social’, ‘B-SSN’), (‘security’, ‘I-SSN’), (‘number’, ‘I-SSN’), (‘(’, ‘O’), (‘SSN’, ‘B-SSN’), (‘:’, ‘O’), (‘123-45-6789’, ‘I-SSN’), (‘)’, ‘O’), (‘and’, ‘O’), (‘credit’, ‘B-CREDIT_CARD’), (‘card’, ‘I-CREDIT_CARD’), (‘number’, ‘O’), (‘(’, ‘O’), (‘CC’, ‘B-CREDIT_CARD’), (‘:’, ‘O’), (‘4111-1111-1111’, ‘I-CREDIT_CARD’), (‘)’, ‘O’), (‘,’, ‘O’), (‘was’, ‘O’), (‘recorded’, ‘O’), (‘and’, ‘O’), (‘inadvertently’, ‘O’), (‘made’, ‘O’), (‘publicly’, ‘O’), (‘available’, ‘O’), (‘.’, ‘O’)]

This example illustrates the annotations necessary for training the Privacy Detection Model. The model consists of two specialized Named Entity Recognition (NER) heads: a Personal NER head and a PII NER head. The Personal NER head is responsible for recognizing personal entity tags such as **{B-PERSON}** and **{I-PERSON}**, thereby identifying entities related to individual identities within the text. In this example, the Personal NER head will specifically learn “John Adams” and the subsequent occurrence of “John” as the personal entities. Note: This version of the model is not trained on the coreferences like ‘she’, ‘her’, ‘his’, ‘he’, ‘their’ etc, due to time constraints.

Simultaneously, the PII NER head is tasked with identifying privacy-sensitive entities such as medical conditions **{B-CONDITION}**, **{I-CONDITION}**, organizational entities like hospital names **{B-ORG}**, **{I-ORG}**, social security numbers **{B-SSN}**, **{I-SSN}**, and credit card details **{B-CREDIT_CARD}**, **{I-CREDIT_CARD}** etc.

Once the Personal and PII NER heads generate their respective embeddings, the model employs a specialized Cross Attention layer. Here, the embeddings from the Personal NER head serve as the Query (Q) vectors, while the embeddings from the PII NER head provide the Key (K) and Value (V) vectors. The attention mechanism thus effectively learns which personal entities (John Adams) are contextually related to sensitive privacy information (hospital admission, drug addiction treatment, SSN, and CC details).

Finally, the output from the Cross Attention layer is aggregated through pooling (such as max pooling), followed by a Feed Forward Network (FFN) and a SoftMax classifier to produce a binary classification decision indicating the presence (label = 1) or absence (label = 0) of privacy-sensitive content.

2.6 Training Strategy

We trained the model in an end-to-end multi-task fashion, optimizing both the NER tagging and the privacy classification objectives simultaneously. The loss function was a weighted sum of three components: the PERSON tagging loss, the PII tagging loss, and the binary classification loss. For tagging, if using a CRF head, the loss is the negative log-likelihood of the correct tag sequence; if using a simpler feed-forward (FF) head, it’s token-level cross-entropy. The classification head uses binary cross-entropy (or equivalently, a sigmoid and cross-entropy loss). We initially set equal weights for all loss components, but also experimented with higher weight on the classification loss to ensure the end goal wasn’t overshadowed by tagging accuracy. In practice, equal or modestly higher weight on classification (e.g., 1.0 for class, 0.5 each for the two tagging losses) worked well to balance learning.

We utilized the ‘AdamW’ optimizer (which is well-suited for transformer training) with a learning rate in the range of 2e-5 to 5e-5 for fine-tuning Modern BERT. A linear learning rate decay schedule with warm-up was employed (warm-up over 5% of steps) to stabilize early training. The model was trained for several epochs, monitoring both tagging F1 and classification accuracy on a validation set.

We used an NVIDIA V100 GPU for training, and the batch size was adjusted (between 16 to 32 sentences per batch) to fit GPU memory, given the model’s multiple heads. Training the multi-task model typically converged in 3–5 epochs on our synthetic dataset (which contained on the order of tens of thousands of sentences).

2.7 Validation Process

We set aside 10% of the synthetic data as a validation set to tune hyperparameters and evaluate generalization. Because the data is synthetic, we also created a small “stress test” set of hand-crafted examples not following the exact templates (to see if the model can handle slight deviations or more complex sentences). During training, we tracked the following metrics:

- NER Tagging F1 Score: for each head (PERSON and PII), computed on the validation set. This measures how well the model identifies the correct entity spans.
- Classification Accuracy and F1: on the validation set, measuring how well it flags sensitive vs. non-sensitive sentences.
- Loss values: total loss and individual components, to ensure all parts are learning (watching for any loss dominating or vanishing).

We observed that the model’s tagging performance usually improves rapidly in early epochs (given the relatively easy synthetic patterns), and the classification performance follows suit. We also ensured that the cross-attention mechanism was indeed being utilized: by inspecting attention weight

distributions, we could verify that in sensitive examples, high attention scores connected person tokens to ID tokens as expected (see Results section for an example). Validation was crucial for deciding some design elements – for instance, whether to include the CRF layer or not, and how much cross-attention helps, which we explore in the ablation study.

3.0 Mathematical Formulation of Model Layers

This section describes the mathematical formulation of the model.

3.1 Transformer Embeddings Layer (Input)

Given a sequence of input tokens

$$X = \{x_1, x_2, \dots, x_n\}$$

the foundational model (e.g., Modern BERT) produces embeddings E :

$$E = \{e(x_1), e(x_2), \dots, e(x_n)\} \quad E \in \{R\}^{n \times d}$$

where:

- n is the sequence length.
- d is the embedding dimension (e.g., 768 for BERT-base).

3.2 Named Entity Recognition (NER) Heads

Personal Head

This head is responsible for tagging tokens that are part of person names. It consists of a linear projection from the BERT hidden state to tag logits for the PERSON entity labels (B-PER, I-PER, O), possibly followed by a CRF layer. In the CRF variant, the CRF layer uses these logits as emission scores and models transition scores between tags, enforcing valid BIO tag sequences (e.g., I-PER must follow B-PER or I-PER, etc.). In a simpler variant, the CRF is omitted, and the linear layer's outputs are normalized with SoftMax for each token. This head outputs a predicted tag sequence for the input.

$$h_{pers} = \text{Dropout}(\text{ReLU}((W_{pers}E + b_{pers})))$$

Followed by either the CRF or a SoftMax Layer

$$Z_{pers} = W_{clspers} h_{pers} + b_{cls_pers}$$

Where:

$$W_{clspers} \in \{R\}^{d' \times d}, d_{clspers} \in \{R\}^{c_{pers}}$$

d' : d expanded by an expansion factor (1.0)

C_{pers} number of personal Entity Tags

PII NER Head: This head identifies other sensitive tokens (we group all other privacy-related entity types into one tag schema for simplicity). It likewise projects the BERT hidden states to logits for PII entity labels (B-PII, I-PII, O), optionally with its own CRF to enforce consistency. This produces a tag sequence like "John/O Doe/O lives/O at/B-PII 123/I-PII Main/I-PII St/I-PII ..." for an address example, or "[SSN]/B-PII [number]/I-PII" for an SSN example.

$$h_{pii} = \text{Dropout}(\text{ReLU}((W_{pii}E + b_{pii})))$$

Followed by either the CRF or a SoftMax Layer

$$Z_{pii} = W_{clspii} h_{pii} + b_{cls_pii}$$

Where:

$$W_{clspii} \in \{R\}^{d' \times d}, d_{clspii} \in \{R\}^{c_{pii}}$$

d' : d expanded by an expansion factor (1.0)

C_{pii} number of PII Entity Tags

These two heads operate on the same BERT output, but they have separate parameters (separate linear/CRF layers). Training them in parallel is a form of multi-task learning on the token level. The intuition is that the PERSON head focusses on human names, while the PII head specializes in detecting sensitive content; each can learn patterns without confusion from the other, unlike a single unified tagger that would have to distinguish many entity types at once.

3.3 Cross-Attention Fusion Layer

After obtaining internal representations from the two NER heads, we introduce a cross-attention layer to fuse their information. In practice, we implement this by taking the final hidden states from the PERSON tagging branch and the PII tagging branch (prior to their classification into tags) and allowing them to attend to each other.

- Queries from Personal NER Embeddings

$$Q = W_Q h_{pers}, W_Q = \{R\}^{d \times d'}$$

- Keys (K) and Values (V) from PII NER Embeddings

$$K = W_K h_{pii}, V = W_V h_{pii}, W_K, W_V = \{R\}^{d \times d'}$$

- then, the scaled dot-product attention is calculated:

$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d'}}\right)V$$

The Final Attended Output is as per the following equation:

$$h_{attn} = \text{Attention}(Q, K, V)$$

3.4 Privacy Classification Head

On top of the cross-attended representations, we place a classification layer to predict if the input text is privacy-sensitive, which pools the attention outputs using max-pooling or in ablation studies using mean pooling. Using max-pooling as an example:

$$h_{pool} = \text{Maxpool}(h_{attn}), h_{attn} \in \{R\}^d$$

followed by a Feedforward with dropout as shown below:

$$h_{cls} = \text{Dropout}(\text{ReLU}((W_{ff}h_{pool} + b_{ff})))$$

Finally, with a classification

$$Z_{cls} = W_{final} h_{cls} + b_{final}$$

3.5 Final Prediction

The final prediction of privacy revealing or not is given by

$$\hat{Y} = \text{Softmax}(Z_{cls})$$

4.0 Synthetic Dataset creation

The synthetic dataset creation steps are outlined in this section.

5.0 Training and Validation

We trained the extended Modern BERT model on the synthetic dataset using the multi-task setup described. The training loop iteratively feeds batches of sentences through the model,

computes losses for each output, and updates weights via gradient descent.

5.1 Training with CRF NER Head

Initial training of the model using CRF NER heads revealed a total negative loss that kept growing epoch after epoch. We observed that the Personal NER head was contributing to the negative loss growth and further investigation revealed that this effect is due to lack of richness of personal tags (B-PER, I-PER) in the input text. Our option was to tag Coreferences like (she, her, his etc..) within the NER tagging. However, we concluded that this is an incorrect approach and decided to change the NER head to SoftMax NER Head.

5.2 Training with SoftMax NER Head

Training with SoftMax gave us good results. The initial training loss curve is shown in Figure 2 that used a dirty dataset. Cleaning the dataset

Figure 2: Modern BERT Losses Unclean data, Expansion=1.0

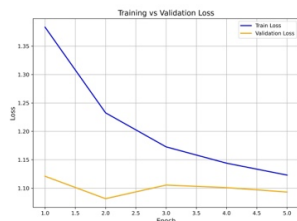
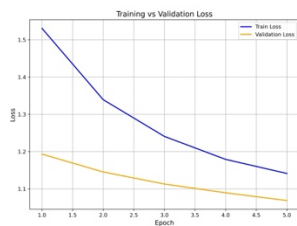
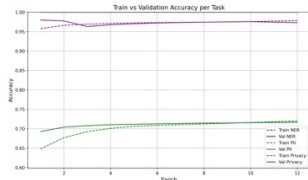


Figure 3: Modern BERT Losses Clean, Expansion=2.0, Dropout=0.25



The accuracy curves are shown in the figure 4:

Figure 4: Modern BERT Accuracy Expansion=2.0, Dropout=0.25



6.0 Ablation Study

To understand the contribution of each component of our architecture, we conducted an ablation study. We created several variant models by removing or altering parts of the full model, then evaluated their performance on the same synthetic test set. The variants included:

- No CRF (Linear Only) for NER Heads: In this variant, we replaced the CRF layers in the NER heads with simple linear classifiers (SoftMax over tags for each token

independently). Everything else remained the same. This is the final model we chose for its stability in training.

- Single NER Head (Unified tags) without Cross-Attention: Here, we tried using a single sequence tagging head to detect all entities (both person names and other PII) combined, essentially treating them as one set of labels (PERSON vs other PII vs none). We did not have enough time to understand the results from these models beneficial.
- No Cross-Attention (Direct Classification): In this variant, we kept the two NER heads but removed the cross-attention layer. ones). We couldn't complete this analysis due to lack of time.
- Alternate Foundational Model: We tried a different model and used "microsoft/deberta-v3-base". This model trained like the Modern Bert.
- Without Person Head or Without PII Head: For completeness, we also tried degenerate cases like training the model with only one of the NER heads (either person or PII) active along with the classifier. These obviously performed poorly on the classification task – if the model only has a person head, it can only tell if a person is present but not if a sensitive token is present, and vice versa. The person-only model flagged almost every sentence with a name as sensitive (high recall, low precision), while the PII-only model flagged any sensitive token presence (missing the link to identity). These variants served as sanity checks that both types of entities are required for accurate detection.

The ablation results are summarized in Table 2. As expected, the full model with two heads + cross-attention + CRFs achieved the best balance, with a privacy classification accuracy of about 98.5% on the test set, whereas removing key components caused drops (e.g., down to ~93% accuracy for the single-head no-attention model). These comparisons validate our design choices: the multiple NER heads provided specialized learning, and the cross-attention provided crucial feature interaction between the heads.

7.0 Results

We evaluated our final model on a held-out test set of synthetic examples (and some manually composed examples for additional verification). Overall, the model's performance was very strong on the synthetic test, which is expected given the training data was similarly distributed. The key results are:

- Privacy Classification: The model achieved 99% accuracy on classifying sentences as sensitive or not. In terms of F1 score for the positive (sensitive) class, it scored around 0.990, indicating almost all privacy-related sentences were correctly identified with very few false alarms or misses. This high performance reflects the relatively clear patterns in the synthetic data. We also evaluated on a small set of more complex sentences (not strictly following templates) and found the accuracy slightly lower (~95%), suggesting that while the model generalizes well, extremely novel phrasing can sometimes confuse it (likely due to distributional differences from training

data).

8.0 Next Steps

This research opens several avenues for future work and improvement. Key next steps include:

- Incorporating Real-World Data in training and prediction.
- Coreference and Context Extension
- Support Encoder-Decoder and Decoder Only Models
- Build Stage 2 of the Model to act on the Results
- Like suppression of Privacy and PII Data
- Impose Differential Privacy
- Expanded Entity Categories
- Model Compression and Deployment
- Evaluation on Diverse Texts: We intend to evaluate the model on different genres of text beyond what it saw in training.
- **Support Privacy Policies.**
- Clinical Notes: Clinical narratives often have PHI that needs removal. Datasets like the i2b2 de-identification challenge set could serve as a benchmark.

9.0 Conclusion

In conclusion, this project demonstrates a successful melding of NER and classification via a transformer to detect privacy-sensitive text. By building on this foundation with the above enhancements, we aim to create a robust, generalizable privacy detection tool. We hope our work contributes to safer data handling practices and inspires further research at the intersection of NLP and privacy.

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